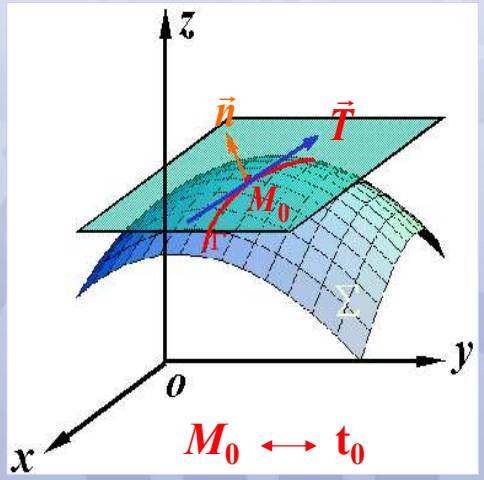
对于光滑曲面$F(x,y,z)=0$,

在$(x_0,y_0,z_0)$处切平面的法向量

$$\vec{n}=\nabla F\Big|_{(x_0,y_0,z_0)}=\{F_x(x_0,y_0,z_0),F_y(x_0,y_0,z_0),F_z(x_0,y_0,z_0)\}$$




在$M_0$处的切平面方程为：

$$F_x(x_0,y_0,z_0)(x-x_0)+F_y(x_0,y_0,z_0)(y-y_0)+F_z(x_0,y_0,z_0)(z-z_0)=0$$

法线方程为：

$$\frac{x-x_0}{F_x(x_0,y_0,z_0)}=\frac{y-y_0}{F_y(x_0,y_0,z_0)}=\frac{z-z_0}{F_z(x_0,y_0,z_0)}$$

设空间曲线Γ：$\begin{cases} F(x,y,z)=0 \\ G(x,y,z)=0 \end{cases}$

$$\vec{\tau} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ F_x & F_y & F_z \\ G_x & G_y & G_z \end{vmatrix}_M$$

例 求曲线 $x^2+y^2+z^2-3z=0, 2x-3y+5z-4=0$

在点(1,1,1)处的切线和法平面.

解:

$$\vec{\tau}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ F_x & F_y & F_z\\ G_x & G_y & G_z\end{vmatrix}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 2x & 2y & 2z-3\\ 2 & -3 & 5\end{vmatrix}$$

在(1,1,1)点解得: $\vec{\tau}=\{7,-12,-10\}$

切线方程 $\dfrac{x-1}{7}=\dfrac{y-1}{-12}=\dfrac{z-1}{-10}$

法平面方程 $7(x-1)-12(y-1)-10(z-1)=0$

换个角度思考：

设空间曲线Γ：$\begin{cases} F(x,y,z)=0 \\ G(x,y,z)=0 \end{cases}$

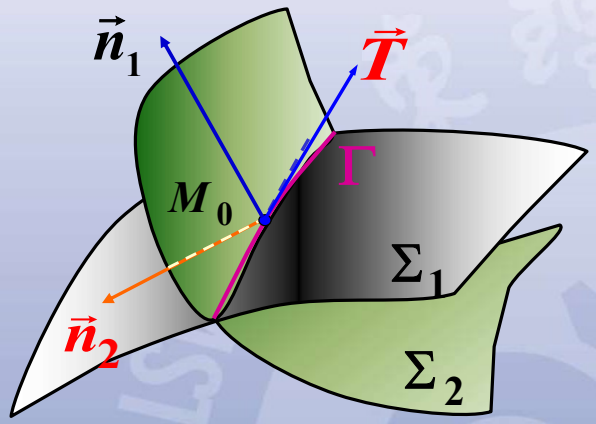
**切平面：切线的集合**

Γ在$M_0$处的切线方程：

$F(x,y,z)=0$在$M_0$处的切平面

$\cap G(x,y,z)=0$在$M_0$处的切平面

$$\vec{\tau}=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ F_x & F_y & F_z \\ G_x & G_y & G_z \end{vmatrix}_M \Rightarrow \text{切向量}=(\nabla F\times\nabla G)\Big|_{M_0}$$



# §11.6 高阶偏导数

$$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) \overset{\Delta}{=} \frac{\partial^2 z}{\partial x^2} = f_{xx}(x,y) = z_{xx} = f_{xx}''$$

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) \overset{\Delta}{=} \frac{\partial^2 z}{\partial x \partial y} = f_{xy}(x,y) = z_{xy} = f_{xy}''$$

$$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) \overset{\Delta}{=} \frac{\partial^2 z}{\partial y \partial x} = f_{yx}(x,y) = z_{yx} = f_{yx}''$$

混合偏导数

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) \overset{\Delta}{=} \frac{\partial^2 z}{\partial y^2} = f_{yy}(x,y) = z_{yy} = f_{yy}''$$

例　设$u = e^{ax}\cos by$，求二阶偏导数.

解　$\dfrac{\partial u}{\partial x} = ae^{ax}\cos by,$　　$\dfrac{\partial u}{\partial y} = -be^{ax}\sin by;$

$$\frac{\partial^2 u}{\partial x^2} = a^2 e^{ax}\cos by, \qquad \frac{\partial^2 u}{\partial y^2} = -b^2 e^{ax}\cos by,$$

$$\frac{\partial^2 u}{\partial x \partial y} = -abe^{ax}\sin by, \qquad \frac{\partial^2 u}{\partial y \partial x} = -abe^{ax}\sin by.$$

**问题：** 混合偏导数都相等吗？

**例** 设 $f(x,y)=\begin{cases}\dfrac{x^3y}{x^2+y^2} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0)\end{cases}$

求 $f(x,y)$的二阶混合偏导数.

**解** 当$(x,y)\neq(0,0)$时,

$$f_x(x,y)=\frac{3x^2y(x^2+y^2)-2x\cdot x^3y}{(x^2+y^2)^2}=\frac{3x^2y}{x^2+y^2}-\frac{2x^4y}{(x^2+y^2)^2},$$

$$f_y(x,y)=\frac{x^3}{x^2+y^2}-\frac{2x^3y^2}{(x^2+y^2)^2},$$

当$(x,y)=(0,0)$时, 按定义可知:

$$f_x(0,0)=\lim_{\Delta x\to 0}\frac{f(\Delta x,0)-f(0,0)}{\Delta x}=\lim_{\Delta x\to 0}\frac{0}{\Delta x}=0,$$

$$f_y(0,0)=\lim_{\Delta y\to 0}\frac{f(0,\Delta y)-f(0,0)}{\Delta y}=\lim_{\Delta y\to 0}\frac{0}{\Delta y}=0,$$

$$f_{xy}(0,0)=\lim_{\Delta y\to 0}\frac{f_x(0,\Delta y)-f_x(0,0)}{\Delta y}=0,$$

$$f_{yx}(0,0)=\lim_{\Delta x\to 0}\frac{f_y(\Delta x,0)-f_y(0,0)}{\Delta x}=1.$$

显然 $f_{xy}(0,0)\neq f_{yx}(0,0).$

定理　(克莱罗定理)

若 $z = f(x, y)$ 的二阶混合偏导数 $f_{xy}, f_{yx}$ 在区域 $D$ 上连续，则在 $D$ 内必有 $f_{xy} = f_{yx}$.

说明　二阶混合偏导数在连续　的条件下，与求导次序 无关.

同样，高阶混合偏导数在偏导 数连续 的条件下，与求导次序 无关.

**例** 设$w = f(x+y+z, xyz)$，$f$具有二阶连续偏导数，求$\dfrac{\partial w}{\partial x}$和$\dfrac{\partial^2 w}{\partial x \partial z}$.

**解** 令 $u = x+y+z$, $v = xyz$;

记 $f_1 = \dfrac{\partial f(u,v)}{\partial u}$, $f_{12} = \dfrac{\partial^2 f(u,v)}{\partial u \partial v}$,

同理有 $f_2$, $f_{21}$, $f_{22}$.

$$\frac{\partial w}{\partial x} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x} = f_1 + yzf_2;$$

$$\frac{\partial^2 w}{\partial x \partial z} = \frac{\partial}{\partial z}(f_1 + yzf_2) = \frac{\partial f_1}{\partial z} + yf_2 + yz\frac{\partial f_2}{\partial z};$$

$$\frac{\partial f_1}{\partial z} = \frac{\partial f_1}{\partial u}\cdot\frac{\partial u}{\partial z} + \frac{\partial f_1}{\partial v}\cdot\frac{\partial v}{\partial z} = f_{11} + xyf_{12};$$

$$\frac{\partial f_2}{\partial z} = \frac{\partial f_2}{\partial u}\cdot\frac{\partial u}{\partial z} + \frac{\partial f_2}{\partial v}\cdot\frac{\partial v}{\partial z} = f_{21} + xyf_{22};$$

于是 $$\frac{\partial^2 w}{\partial x \partial z} = f_{11} + xyf_{12} + yf_2 + yz(f_{21} + xyf_{22})$$

$$= f_{11} + y(x+z)f_{12} + xy^2 zf_{22} + yf_2.$$

**例** 设 $z = f\left(\frac{y}{x}\right)$，$f(u)$有二阶导数，求$\frac{\partial^2 z}{\partial x^2}$、$\frac{\partial^2 z}{\partial x \partial y}$.

**解**

$$\frac{\partial z}{\partial x} = f'\left(\frac{y}{x}\right)\cdot\left(-\frac{y}{x^2}\right)$$

$$\frac{\partial^2 z}{\partial x^2} = f''\left(\frac{y}{x}\right)\cdot\left(\frac{y^2}{x^4}\right) + f'\left(\frac{y}{x}\right)\cdot\left(\frac{2y}{x^3}\right)$$

$$\frac{\partial^2 z}{\partial x \partial y} = f''\left(\frac{y}{x}\right)\cdot\left(-\frac{y}{x^3}\right) + f'\left(\frac{y}{x}\right)\cdot\left(-\frac{1}{x^2}\right)$$

例 已知$u = f(2x+3y)+\varphi(2x,3y)$. 求$u_{xy}$。

解: $u_x = f'(2x+3y)\cdot 2+\varphi_1(2x,3y)\cdot 2$

$$u_{xy} = 6f''(2x+3y)+6\varphi_{12}(2x,3y)$$

例　设$x^2+y^2+z^2-4z=0$，求$\dfrac{\partial^2 z}{\partial x^2}$.

解　令　$F(x,y,z)=x^2+y^2+z^2-4z,$

则　$F_x=2x,\ F_z=2z-4,\quad \dfrac{\partial z}{\partial x}=-\dfrac{F_x}{F_z}=\dfrac{x}{2-z},$

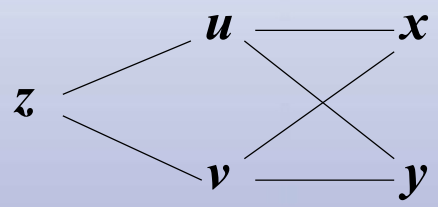
$$\frac{\partial^2 z}{\partial x^2}=\frac{(2-z)+x\dfrac{\partial z}{\partial x}}{(2-z)^2}=\frac{(2-z)+x\cdot\dfrac{x}{2-z}}{(2-z)^2}$$

$$=\frac{(2-z)^2+x^2}{(2-z)^3}.$$

例　方程$\dfrac{\partial z}{\partial x}=3\dfrac{\partial z}{\partial y}$　在变换$u=x+3y, v=3x+y$

下，可得的新方程为什么？

分析：本题即是将方程换元，函数不变，自变量为$u,v$.




解：$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x}=\frac{\partial z}{\partial u}+3\frac{\partial z}{\partial v}$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial y}=3\frac{\partial z}{\partial u}+\frac{\partial z}{\partial v}$$

代入 方程$\frac{\partial z}{\partial x}=3\frac{\partial z}{\partial y}$得，

$$\frac{\partial z}{\partial u}+3\frac{\partial z}{\partial v}=3(3\frac{\partial z}{\partial u}+\frac{\partial z}{\partial v})$$

化简

$$\frac{\partial z}{\partial u}=0$$